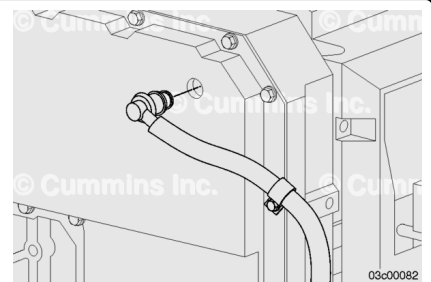


## Trainings ▾

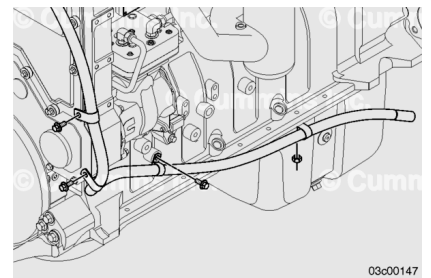
### Remove

#### Automotive with CM870

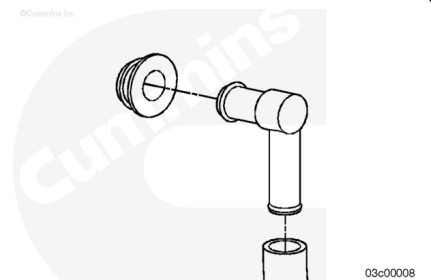
Remove the breather tube and rubber grommet from the engine.



Remove the four p-clips securing the hose to the front cover and block.

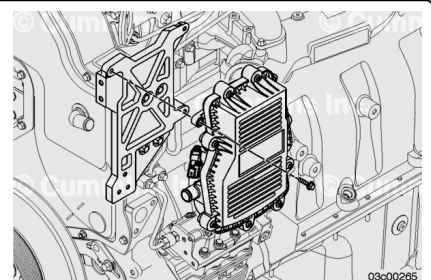


Disassemble the breather tube.



#### Automotive With CM871

Some engines are equipped with a crankcase breather heater and insulation option. If the engine is equipped with the crankcase breather heater, disconnect the heater wire harness from the engine wire harness. The connection is located at the bottom left of the crankcase breather assembly.



Disconnect the wiring connector from the breather pressure sensor.

Loosen the two hose clamps on the inlet hose.

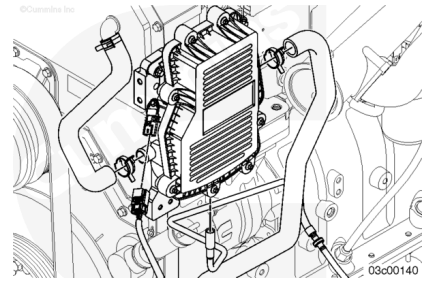
Disconnect the hose from breather. Loosen the outlet hose clamp.

Disconnect the outlet hose.

Disconnect the breather oil drain line.

Remove the drain line inline with the connection nipple. Do **not** bend.

Remove the four mounting capscrews from the crankcase breather element housing to the breather housing bracket and remove the housing assembly.

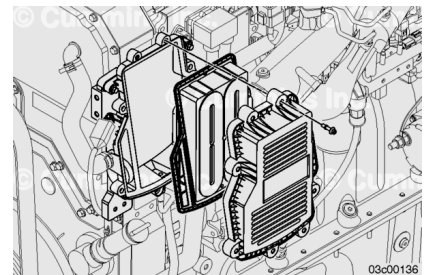


## Disassemble

### Automotive With CM871

Disassemble the crankcase breather assembly. Refer to Procedure 003-019 in Section 3 (</qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html>).

**Note :** All capscrews are captive capscrews.

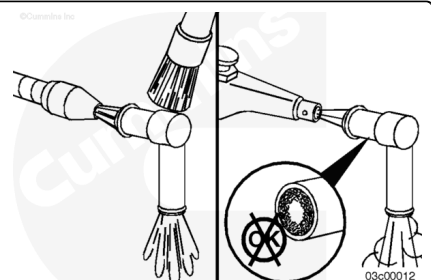


## Clean and Inspect for Reuse

### Automotive with CM870

#### **⚠ WARNING ⚠**

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Use solvent to clean the inside of the crankcase breather tube, and dry with compressed air.

Use air pressure to blow through the vent tube.

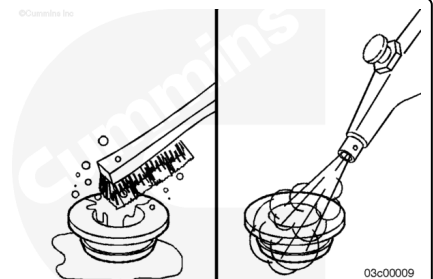
Replace the vent tube if it is clogged.

**⚠ WARNING ⚠**

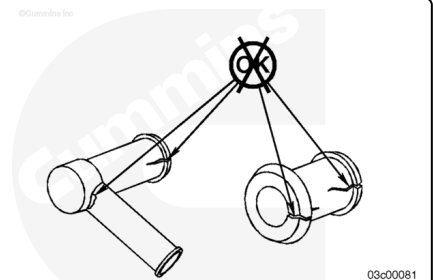
**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Do **not** use solvent to clean the rubber grommet.

Clean the rubber grommet with soapy water. Dry with compressed air.



Inspect the breather tube, hose and rubber grommet for cracks or other damage. Replace if necessary.



## Automotive With CM871

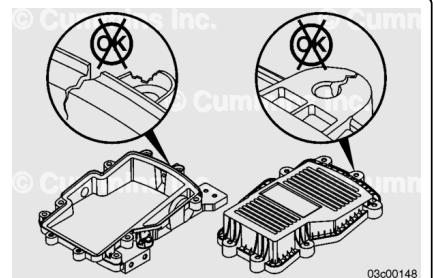
**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.**

Clean the breather crankcase housing and breather crankcase cover with solvent.

Dry with compressed air.

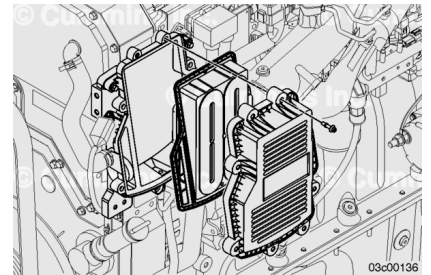
Check for cracks in the crankcase breather element housing and crankcase breather cover. Replace if necessary.



## Assemble

### Automotive With CM871

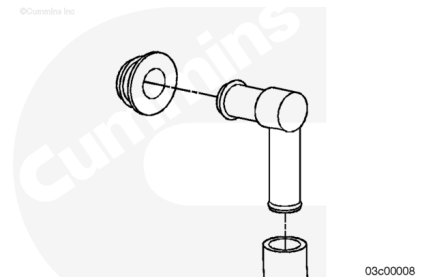
Assemble the crankcase breather assembly. Refer to Procedure 003-019 in Section 3 ([/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html](http://qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)).



## Install

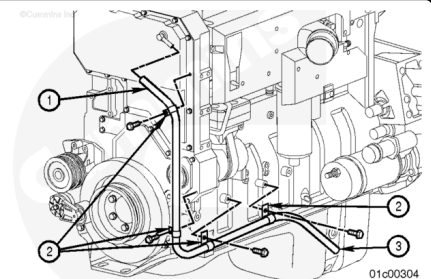
### Automotive with CM870

Assemble the breather tube as shown.



Two p-clips attach the vent hose to the gear housing and two p-clips attach the vent hose to the side of the block. With the breather hose installed on the new cover, the hose (3) **must** extend lower than the flywheel housing.

Install the crankcase vent hose (1). Use the four p-clips (2), to secure the crankcase vent hose to the engine.



**Torque Value:** 28 n•m [ 21 ft-lb ]

### Automotive With CM871

Connect the inlet and outlet rubber hoses, oil drain line, and install the three hose clamps to the housing. Be sure to push the hose on until it stops on the housing connector. Verify that the hose clears the fan blade.

Connect the breather sensor wiring connector to the breather sensor.

Install and tighten the four mounting capscrews.

**Torque Value:**

M10

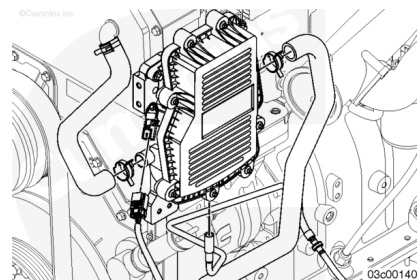
1. 47 n•m [ 35 ft-lb ]

**Torque Value:**

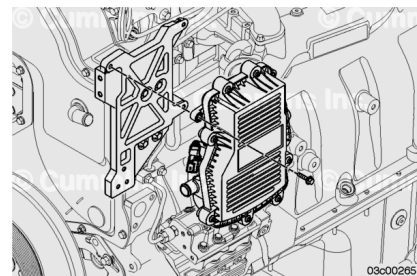
M8

1. 20 n•m [ 177 in-lb ]

Remove and install the crankcase pressure sensor. Use the following procedure in the ISX CM871 and ISM CM876 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-445 in Section 19. (</qs3/pubsys2/xml/en/procedures/00/00-019-445.html>)



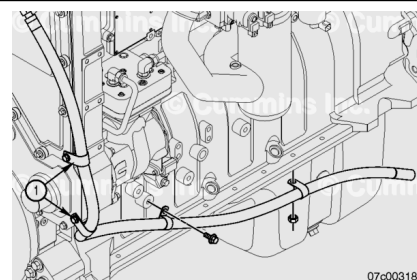
Some engines are equipped with a crankcase breather heater and insulation option. If the engine is equipped with the crankcase breather heater, connect the heater wire harness from the engine wire harness. The connection is located at the bottom left of the crankcase breather assembly.



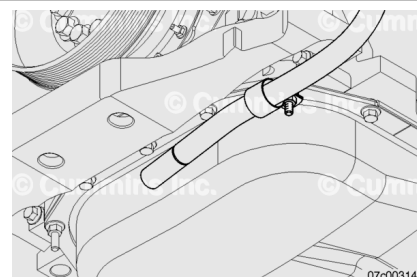
## Modify

Be sure to use these two p-clips (1) with the new crankcase vent hose routing.

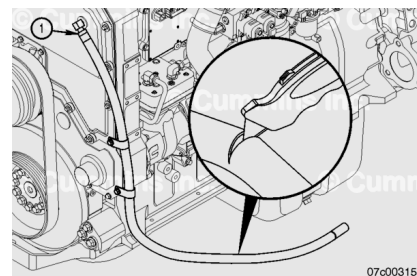
Remove any other p-clips found.



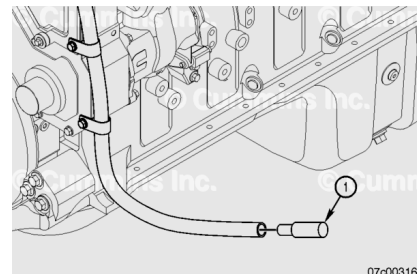
The new crankcase vent hose routing runs in front of the oil pan and ends below the pan flange.



Cut the existing crankcase vent hose to a length of 1.02 m [40 in] measuring from the top (1).



Remove the plastic piece (1), Part Number 3819475, from the end of the old crankcase vent hose and install on the end of the modified crankcase vent hose.



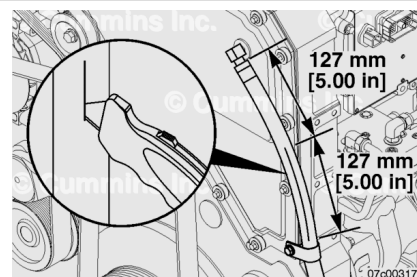
From the top of the hose, measure down 127 mm [5 in] and start the slit cut in the hose.

The slit should be in the center of the hose.

Measure down 254 mm [10 in] and stop the slit. This will make the slit 127 mm [5 in] long.

Make the cut as smooth as possible.

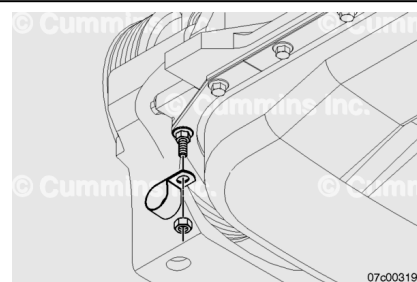
The slit should be in the center of the hose, through one wall.



Install a studded capscrew, Part Number 4026812, in this location.

This stud may be taken from a location where it is **not** being used.

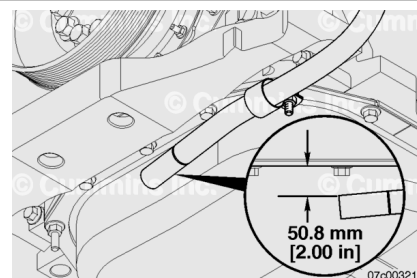
Attach a p-clip, Part Number 107460, to the new studded capscrew in this position using an M8 x 1.25 nut.



### **⚠ CAUTION ⚠**

**The crankcase vent hose must never have a built-in trap which allows fluid to form in the tube.**

The plastic piece at the end of the crankcase vent hose should touch the pan as much as possible.



The plastic end of the crankcase vent hose should be near the middle of the oil pan.

The crankcase vent hose **must** slope and end 50 mm [2 in] below the pan rail.

Rear Sump Oil Pan Application: Follow the same procedures as outlined for the front sump application.

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**Last Modified: 24-Apr-2015**

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